

THE INTEGRAL SIGNED COVER OF AN ORIGINAL ERDŐS–STRAUS STATE

THE CLANKERS

ABSTRACT. The literal-root quartic supplied with the ES–Zeta continuation has a finite flat signed completion of rank eight. At an original prime we compute its integral normalization, not merely its geometric degree. For every original state at $p \equiv 1 \pmod{12}$, its normalization index is p^2 in Type I and p^9 in Type II, with explicit conductors, invariant factors, root labels and inverse lattices. The added Type-I boundary point lifts modulo p^2 but never modulo p^3 . We also give an explicit joint-channel finite-place test family: both numerical gates and the displayed height bounds hold, but one prime occurrence violates the original divisor box. The two rational targets retain their exact prime-local E/M signatures. The same primes have separately exhibited integer ES solutions. These are local and integral comparison theorems, not a proof of universal ES occupancy.

The original integral signed order and its exact normalization

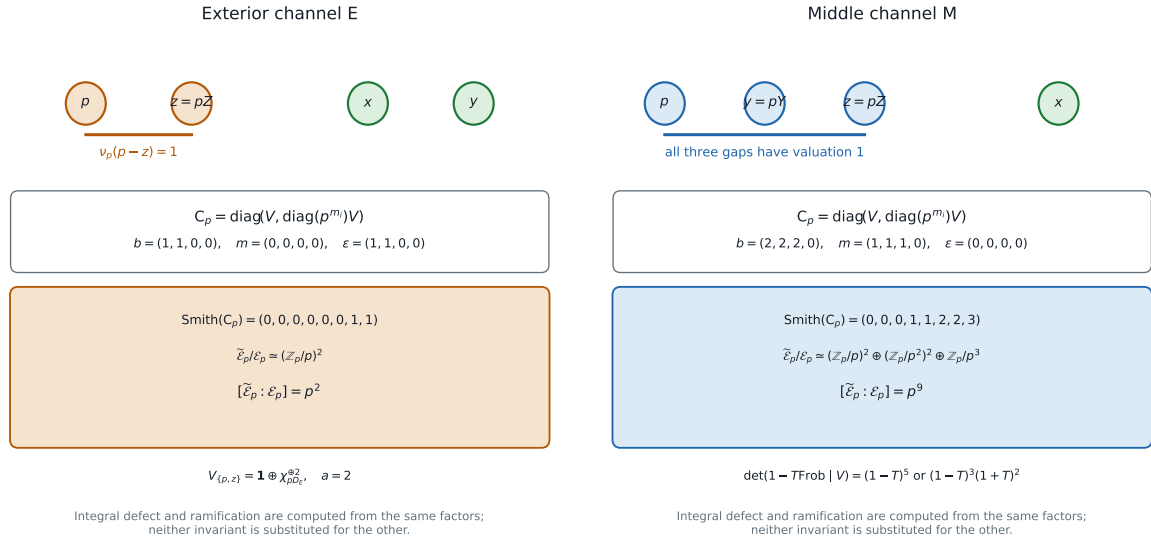


FIGURE 1. The exact Type-I/Type-II normalization matrix and its Smith factors in the retained original basis. Finite rank stays eight while the integral lattice indices become p^2 and p^9 .

1. QUESTION, INHERITED OBJECTS, AND THE UNRESOLVED IMPLICATION

The preceding capacity theorem [7] completes a fixed-target divisor problem. It proves that searching additional divisors at a fixed grade cannot repair every exterior source. Its universal existence obligation is still open: for a prescribed hard prime, produce an original exterior or middle state. We do not assume that obligation in the present calculation.

The newly supplied ES–Zeta work [5, §§1–3, equations (8)–(24)] uses the *literal* four roots (p, x, y, z) and completes its signed inverse by $\eta = \xi^{-1}$. Its monic equations give a finite flat rank-eight algebra. The task here is to determine what happens to that particular algebra over

$\mathbb{Z}_p$, with its original coordinate lattice, rather than replacing it by an arbitrary rank-eight vector space. The literal quartic is different from the reciprocal cubic in [6]; the exact inversion and its derivative twist are given in Section 7.1.

The Type-I/II identities are classical [1, §2, Propositions 2.2 and 2.6]. Quadratic character restrictions are also antecedents; in particular compare [2, Corollary 1.3 and Appendix A]. All calculations needed for the local and integral results below are proved explicitly. No historical-priority claim is made for these specializations or for the standard local algebra tools.

2. ORIGINAL COORDINATES AND THE LITERAL POLYNOMIAL

Fix a prime $p \equiv 1 \pmod{12}$. Thus $p \geq 13$, and both -1 and 3 are squares modulo p . Set $K = \mathbb{Q}_p$, $\mathfrak{o} = \mathbb{Z}_p$, and normalize $v_p(p) = 1$. An original state retains

$$(IC1) \quad p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad E : R \mid 4u + 1, \quad M : R \mid u + a.$$

For $a = \prod_l l^{e_l}$ its original exponent vector is $\beta_l = v_l(u) - e_l \in [-e_l, e_l]$. Let

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g).$$

Then $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$. The exact ordered returns are

$$(IC2) \quad \begin{aligned} E : \quad \kappa &= (pr + s)/R, & (x, y, z) &= (a, hsk, phr\kappa), \\ M : \quad \lambda &= (r + s)/R, & (x, y, z) &= (a, phs\lambda, phr\lambda). \end{aligned}$$

Multiplication proves $4/p = 1/x + 1/y + 1/z$. The ordered divisor inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$, respectively. The middle complement $u \mapsto a^2/u$ exchanges the last two denominators and is kept as an orientation, not suppressed as a duplicate.

Lemma 2.1 (Original character and unit facts). *Every factor in (IC2) is a p -unit. For each $d \mid a^2$, $\left(\frac{d}{R}\right) = \left(\frac{d}{p}\right)$, with Jacobi symbol on the left. Consequently*

$$(IC3) \quad E : \left(\frac{h}{p}\right) = -1, \quad M : \left(\frac{rs}{p}\right) = -1.$$

In the exterior case, $\kappa - r$ and $\kappa + r$ are p -units.

Proof. For an odd $l \mid a$, reciprocity and $R \equiv -p \pmod{l}$ give $\left(\frac{l}{R}\right) = \left(\frac{-1}{l}\right) \left(\frac{R}{l}\right) = \left(\frac{p}{l}\right) = \left(\frac{l}{p}\right)$. If $2 \mid a$, use $R \equiv -p \pmod{8}$ and the supplementary law. Multiplication over the original exponents proves the divisor identity. The E target is $-1/4$ modulo $R \equiv 3 \pmod{4}$, so $\left(\frac{u}{p}\right) = -1$ and $u = hr^2$ gives the first assertion. The M relation $r \equiv -s \pmod{R}$ gives the second.

The factors h, r, s are below p . In E put $D = (4u + 1)/R = (r + \kappa)/s$; the original identities give $pD + 1 = 4hr\kappa$, so κ is a unit too. If $p \mid \kappa - r$, this identity gives $4hr^2 \equiv 1 \pmod{p}$, contradicting $\left(\frac{h}{p}\right) = -1$. If $p \mid \kappa + r$, then $p \mid D$ and $4u + 1 = RD$ gives the equally impossible nonresidue $u \equiv -1/4$. For M, $r + s \leq rs + 1 \leq a + 1 < p$, so $\lambda < p$. $\square$

The fixed, ordered root vector and scale are

$$(\ell_0, \ell_1, \ell_2, \ell_3) = (p, x, y, z), \quad S = p + x + y + z,$$

$$(IC4) \quad \mathcal{H}(T) = -S^{-1} \prod_{i=0}^3 (T - \ell_i) = AT^4 + T^3 + BT^2 + CT + D.$$

Here D is a quartic coefficient, not the original exterior cofactor D . Vieta and the reciprocal identity give

$$(IC5) \quad A = -S^{-1}, \quad C = 5xyz/S, \quad D = -pxyz/S, \quad D = -pC/5, \quad B = -Ap^2 - p - 4C/(5p).$$

These are the literal coefficient equations in [5, §2.1]; the prime and every root label are retained. Define $d_i = \mathcal{H}'(\ell_i)$.

Theorem 2.2 (Original prime-local root clusters). *The roots are distinct, $S \in \mathfrak{o}^\times$, and the following assertions hold. For E write $(x, y, z) = (a, b, pc)$. Then $a, b, a - b, a + b$ are units, $c \equiv 1/4 \pmod{p}$, and*

$$(IC6) \quad (v_p(d_0), v_p(d_1), v_p(d_2), v_p(d_3)) = (1, 0, 0, 1), \quad v_p \text{Disc}(\mathcal{H}) = 2.$$

For M write $(x, y, z) = (a, pb, pc)$. Then $b, c, b - c, b - 1, c - 1$ are units,

$$4bc \equiv b + c \pmod{p}, \quad \left(\frac{bc}{p}\right) = -1,$$

$$(IC7) \quad (v_p(d_0), v_p(d_1), v_p(d_2), v_p(d_3)) = (2, 0, 2, 2), \quad v_p \text{Disc}(\mathcal{H}) = 6.$$

In coefficient order (A, B, C, D) the valuations are respectively $(0, 0, 1, 2)$ and $(0, 1, 2, 3)$.

Proof. For E, $c = hr\kappa$ and $4c = pD + 1$. The remaining unit statements follow from Lemma 2.1 and $a \pm b = hs(r \pm \kappa)$. Thus exactly one pair of roots, p and pc , has difference of valuation one; all other differences are units. In particular they are all distinct, and $S \equiv a + b$ is a unit. The derivative and discriminant formulas prove (IC6). The coefficient $B \equiv -ab/S$ is a unit, and (IC5) gives the other valuations.

For M, $b = hs\lambda, c = hr\lambda$. The unit and distinctness of b, c follow from $r \neq s, r, s < p$; equality would force $r = s = 1$ and $R \mid 2$. The ES equation gives $4bc = pbc/a + b + c$, and (IC3) gives $\left(\frac{bc}{p}\right) = -1$. If $b = 1 \pmod{p}$, then $c = 1/3$, making bc a square since $\left(\frac{3}{p}\right) = 1$. The same excludes $c = 1$. The three roots p, pb, pc therefore have pairwise valuation-one differences; the singleton a is separated by units. Now $S \equiv a$, giving (IC7). In B the leading term is $-pa(1 + b + c)/S$, and $1 + b + c \equiv 1 + 4bc$ cannot vanish, since $bc = -1/4$ would be a square. Hence $v_p(B) = 1$. The remaining values follow from (IC5). $\square$

In particular any two original E and M coefficient targets at the same prime have distance exactly one in the maximum p -adic norm on (A, B, C, D) : the B -coordinate is a unit on E and divisible by p on M, while all coordinates are integral. An infinitesimal target deformation preserving reduction is not such a channel change. No comparison with a complex Gamma metric is inferred.

3. THE RANK-EIGHT GENERIC ALGEBRA AND ITS ORIGINAL INTEGRAL ORDER

The supplied completion specializes, with its exact original basis, to

$$(IC8) \quad \mathcal{O} = \mathfrak{o}[T]/(\mathcal{H}), \quad \widehat{\mathcal{S}} = \mathcal{O}[\eta]/(\eta^2 - \mathcal{H}'(T)).$$

The leading coefficient A is a unit, so both equations can be made monic. Thus $\widehat{\mathcal{S}}$ is free over $\mathfrak{o}$ on

$$(IC9) \quad 1, T, T^2, T^3, \eta, \eta T, \eta T^2, \eta T^3.$$

After inverting η it is the original signed inverse, with $\xi = \eta^{-1}$ and $\xi^2 \mathcal{H}'(T) = 1$. The full physical Fabel coordinates, inherited from [5, equation (21)], are

$$(IC10) \quad \left(\eta^{-1}, -T - i\eta, A\eta^3 + 2T\eta + 3i\eta^2, 7iT^2\eta + (B - 17T + AT^2)\eta^2 - 13i\eta^3 - 2A\eta^4 \right).$$

Either embedding of i in K may be chosen and is kept fixed. No complex period coefficient is presumed to have a p -adic descent.

Over K , root evaluation gives exactly

$$(IC11) \quad \widehat{\mathcal{S}}_K \simeq \prod_{i=0}^3 K[\eta_i]/(\eta_i^2 - d_i).$$

There is no infinity factor: $A \neq 0$ and all four roots are finite. This is not the seven-dimensional reciprocal-cubic algebra.

Theorem 3.1 (Generic local signed types). *An E state has either*

$$(IC12) \quad K^4 \times K_{\text{ram}}^2 \quad \text{or} \quad K_{\text{unr}}^2 \times K_{\text{ram}}^2,$$

and consequently four or zero K-rational signed inverse points. An M state has either

$$(IC13) \quad K^8 \quad \text{or} \quad K^4 \times K_{\text{unr}}^2,$$

and consequently eight or four such points. Repeated field factors retain their root labels; they are not identified.

Proof. For E the two valuation-one derivatives have square ratio: after dividing by p their ratio is $-1 \pmod{p}$. They define the same ramified quadratic extension. At the other two roots the derivative residues are

$$-\frac{a^2(a-b)}{a+b}, \quad -\frac{b^2(b-a)}{a+b}.$$

Their ratio is a square, so either both split or both define the unramified quadratic extension. This proves (IC12).

For M, divide the three cluster derivatives by p^2 . Their unit residues are

$$(IC14) \quad (1-b)(1-c), \quad (b-1)(b-c), \quad (c-1)(c-b).$$

Their product is the negative of a nonzero square. Since -1 is a square, either all three or exactly one are squares. At the singleton, $d_1 \equiv -a^2$ is a square. This proves (IC13).

For the local facts, a square unit modulo the odd prime has unique compatible lifts after a sign is chosen: if $z_n^2 \equiv d \pmod{p^n}$, set $z_{n+1} = z_n + p^n k$, with $k \equiv -(z_n^2 - d)/p^n (2z_n)^{-1} \pmod{p}$. A nonsquare unit has irreducible separable quadratic reduction and gives the unramified quadratic extension. An odd-valuation class gives a ramified quadratic. These are also the standard constructions in [3, Chapter 7, Newton's lemma and local extensions]. $\square$

At the single hard prime $p = 1009$, all four alternatives occur. The respective original (a, u) and ordered denominators are

	channel, number of K -points	(a, u)	(x, y, z)
(IC15)	$E, 0$	(253, 11)	(253, 87032, 3818056)
	$E, 4$	(253, 23)	(253, 86020, 7890380)
	$M, 8$	(253, 11)	(253, 2042216, 88792)
	$M, 4$	(253, 23)	(253, 1021108, 92828).

These are four independently checked original states. In particular a K -rational point on the literal signed fibre is not a necessary condition for an ES witness. Every row still has eight geometric characteristic-zero points.

4. EXACT NORMALIZATION, CONDUCTORS AND INVARIANT FACTORS

Put $k_i = \lfloor v_p(d_i)/2 \rfloor$ and

$$(IC16) \quad \theta_i = \eta_i/p^{k_i}, \quad \delta_i = d_i/p^{2k_i}, \quad N_i = \mathfrak{o}[\theta_i]/(\theta_i^2 - \delta_i), \quad \mathcal{N} = \prod_i N_i.$$

Each δ_i is a unit or has valuation one. Thus N_i is the maximal integral order in its generic factor. For a field factor, integrality of $a + b\theta_i$ first gives $a \in \mathfrak{o}$ by its trace. Its norm then gives $2v_p(b) + v_p(\delta_i) \geq 0$, forcing $b \in \mathfrak{o}$ because $v_p(b)$ is integral. For a split unit factor use the two values $a \pm b\sqrt{\delta_i}$ and the unit $2\sqrt{\delta_i}$. This also proves maximality without an unspecified integral basis.

The precise normalization map is

$$(IC17) \quad f(T) + \eta g(T) \mapsto (f(\ell_i) + p^{k_i} g(\ell_i) \theta_i)_{i=0}^3, \quad \deg f, \deg g < 4.$$

It is injective: over K , evaluate at all distinct labelled roots. All its inverse coefficients are provided by the corresponding Lagrange polynomials, so its integral image can be tested without changing the lattice.

Theorem 4.1 (Complete original integral defect). *For E,*

$$(IC18) \quad \mathcal{N}/\widehat{\mathcal{S}} \simeq (\mathfrak{o}/p)^2, \quad [\mathcal{N} : \widehat{\mathcal{S}}] = p^2, \quad \mathfrak{f}_E = pN_0 \times N_1 \times N_2 \times pN_3.$$

For M,

$$(IC19) \quad \mathcal{N}/\widehat{\mathcal{S}} \simeq (\mathfrak{o}/p)^2 \oplus (\mathfrak{o}/p^2)^2 \oplus \mathfrak{o}/p^3, \quad [\mathcal{N} : \widehat{\mathcal{S}}] = p^9, \\ \mathfrak{f}_M = p^3N_0 \times N_1 \times p^3N_2 \times p^3N_3.$$

Here $\mathfrak{f} = \{n \in \mathcal{N} : n\mathcal{N} \subseteq \widehat{\mathcal{S}}\}$ is the conductor ideal in this finite algebra. It is not the analytic weighted conductor of the Zeta programme. In the eight-dimensional original basis, the Smith exponents are

$$(IC20) \quad E : (0, 0, 0, 0, 0, 0, 1, 1), \quad M : (0, 0, 0, 1, 1, 2, 2, 3).$$

Proof. Unit differences split off every noncolliding root integrally, by the polynomial CRT. For E, the remaining root lattice on the pair p, pc is exactly

$$L_2 = \{(z_0, z_3) \in \mathfrak{o}^2 : z_0 - z_3 \in p\mathfrak{o}\}.$$

A linear interpolation proves sufficiency, since $p - pc$ is p times a unit. Both coefficients f and g in (IC17) give a copy of L_2 : here $k_i = 0$. This proves the quotient and exponents. A normal element supported at just one of these two roots is in the original order precisely when its coefficients of 1 and θ_i both lie in $p\mathfrak{o}$. The resulting pN_i is an ideal and is maximal with this property, giving (IC18).

For M, order the cluster as p, pb, pc and use the Newton polynomial basis $1, T - p, (T - p)(T - pb)$. Evaluation gives diagonal factors $1, p, p^2$ times units, and integral row operations remove the lower entries. Thus its root lattice L_3 has quotient $\mathfrak{o}/p \oplus \mathfrak{o}/p^2$. The two coefficients in (IC17) give $L_3 \oplus pL_3$. Their exponents are $(0, 1, 2)$ and $(1, 2, 3)$; the singleton contributes two zeros. This proves (IC19)–(IC20).

For the conductor, a vector supported at one of the three roots has a unique Lagrange interpolation of degree at most two. Its leading coefficient has denominator of exact order p^2 . Hence a normal value $a + b\theta_i$ supported there belongs to $\widehat{\mathcal{S}}$ precisely when $a \in p^2\mathfrak{o}$ and $b \in p^3\mathfrak{o}$. For it to lie in the conductor, multiplication by the unit θ_i must also belong to the order. This forces $a, b \in p^3\mathfrak{o}$, and that condition is sufficient because $1, \theta_i$ is the full normal basis. The singleton is already maximal. This proves the displayed conductor. $\square$

The root orders alone have indices p (E) and p^3 (M), with Smith exponents $(0, 0, 0, 1)$ and $(0, 0, 1, 2)$. A further exact check is the trace discriminant:

$$v_p \text{Disc}(\widehat{\mathcal{S}}) = 2v_p \text{Disc}(\mathcal{O}) + \sum_i v_p(4d_i) = 6 \text{ (E)}, \quad 18 \text{ (M)}.$$

The normal algebra has discriminant valuation two (E) or zero (M). The square of the change-of-basis determinant again gives index exponents two and nine. Neither determinant is a density or positivity assertion.

Corollary 4.2 (The original actions on the finite defect module). *Let $Q = \mathcal{N}/\widehat{\mathcal{S}}$, a module rather than a quotient ring. Multiplication by the original T and η descends to it. Their exact nilpotence indices are*

$$(IC20a) \quad E : (1, 2), \quad M : (3, 3),$$

respectively. Thus the finite quotient retains nonzero action, even though rational extension removes it.

Proof. The quotient is supported only at the colliding roots. For E, T is divisible by p there and $\eta^2 = d_i$ is also divisible by p , so the conductor kills both actions. Multiplying a supported normal constant by η still has a coefficient not divisible by p and is nonzero in Q . For M, both T^3 and η^3 have a factor p^3 at each colliding normal factor. But their squares, applied to a supported normal θ_i , have a θ_i -coefficient of exact valuation two. The membership criterion in the proof of Theorem 4.1 requires three, so both squared actions remain nonzero. All the normal root quotients are units, precluding cancellation. $\square$

Complete inverse coordinates. For a normal vector $a_i + b_i\theta_i$, interpolate the four values a_i to obtain $f(T)$, and the values b_i/p^{k_i} to obtain $g(T)$. Membership in the original order means all eight coefficients of f, g are in $\mathfrak{o}$. Thus (IC17) has an explicit partial integral inverse and a total K -linear inverse. The quotient in (IC18) or (IC19) is retained when the integral inverse fails. There is no $\mathfrak{o}$ -linear splitting of its surjection from $\mathcal{N}$, since that would map a nonzero torsion module into a torsion-free one.

5. BOUNDARY FIBRES AND A SHARP FINITE LIFTING OBSTRUCTION

At E, the colliding factor of the special fibre is

$$(IC21) \quad \mathbb{F}_p[\epsilon, \eta]/(\epsilon^2, \eta^2 - 2g_0\epsilon) \simeq \mathbb{F}_p[\eta]/(\eta^4), \quad g_0 = -ab/S \neq 0.$$

At M it is

$$(IC22) \quad \boxed{\mathbb{F}_p[\epsilon, \eta]/(\epsilon^3, \eta^2 - 3\epsilon^2)}.$$

These follow from the actual reduced quartics, respectively $-S^{-1}T^2(T-a)(T-b)$ and $T^3(1-T/a)$, using the supplied multiple-root algebra [5, equation (23)]. Their dimensions are four and six. Multiplication by ϵ has blocks (2, 2) or (3, 3). Multiplication by η has blocks (4) or (4, 2): on $1, \epsilon, \epsilon^2, \eta, \epsilon\eta, \epsilon^2\eta$ in (IC22), its successive ranks are 6, 4, 2, 1, 0. Thus (IC22) is not a single length-six chain.

The normalization map modulo p has kernel dimension two on the E collision block and five on the M collision block. For E, T becomes zero in each normal factor modulo p , while η gives its two labelled nonzero nilpotents. For M, both $T = p\zeta$ and $\eta = p\theta$ become zero in the normal special fibre. These are the kernels dictated by (IC20), not a deletion of the original labels.

Theorem 5.1 (The added E point stops at the third digit). *The point $(T, \eta) = (0, 0)$ of (IC21) has exactly p lifts solving both original equations modulo p^2 , and has no lift modulo p^3 .*

Proof. Write $\mathcal{H}(T) = U(T)(T-p)(T-pc)$, with $U(T) = -(T-a)(T-b)/S$ a unit near zero. A putative lift modulo p^2 is $T = pw, \eta = pv$, with $w, v \in \mathbb{F}_p$. The first equation is automatic. The second gives

$$0 \equiv pU(0)(2w - 1 - c) \pmod{p^2},$$

so $w = (1+c)/2 = 5/8 \pmod{p}$. The p values of v are all possible. For any higher lift of that T ,

$$(IC23) \quad \frac{\mathcal{H}(T)}{p^2} \equiv -U(0)\frac{(1-c)^2}{4} = \frac{9ab}{64S} \not\equiv 0 \pmod{p}.$$

Thus the root equation already fails modulo p^3 , independently of η . $\square$

The finite flat completion is proper, but this boundary point has no integral $\mathfrak{o}$ -section. The ramified generic points remain over their actual quadratic extensions. At $p = 1009$ in the first row of (IC15), the entire generic signed fibre has no K -point despite an original ES solution and a rational special boundary point. No smooth Hensel hypothesis is available at $\eta = 0$.

For M, the collision block instead has the exact integral normalization chart

$$(IC24) \quad G(Z) = (Z-1)(Z-b)(Z-c), \quad G(Z) = 0, \quad \theta^2 = \frac{a-pZ}{S}G'(Z),$$

with $T = pZ, \eta = p\theta$. This is finite etale of rank six, since the reduced roots are distinct and the right side is a unit there. The inverse divides exactly by those powers of p . The character product in (IC14) proves that at least one cluster root has two K -rational signed lifts. Thus the M boundary point does lift to $\mathfrak{o}$, although its full multiplicity and the index p^9 are still retained.

The original physical inverse (IC10) has pole $v_p(\xi) = -1/2$ on the E cluster and $v_p(\xi) = -1$ on the M cluster. Inverting η in (IC21) or (IC22) annihilates that local algebra; it does not turn the added state into a finite original affine inverse. The supplied complex metric limit remains the one in [5, equations (21)–(22)]; no identification of that complex norm with this nonarchimedean valuation is asserted.

6. THE GENERIC PRIME-SIGN SYMMETRY HAS AN INTEGRAL DEFECT

The distinguished-root projector is

$$(IC25) \quad e_p(T) = \frac{\prod_{i=1}^3 (T - \ell_i)}{\prod_{i=1}^3 (p - \ell_i)}.$$

Its values are $(1, 0, 0, 0)$. Its coefficients require an exact denominator p in E and p^2 in M , since its numerator is monic and the denominator has those respective valuations. This is also the supplied g_p/d_p projector [5, equations (9)–(11)].

Among the supplied four generic deck sign maps, the prime-only flip is

$$(IC26) \quad T \mapsto T, \quad \eta \mapsto (1 - 2e_p(T))\eta.$$

Uniqueness in the original free basis (IC9) proves that it does not preserve $\hat{S}$. Nor does its negative, the denominator-only flip. The identity and the simultaneous sign $\eta \mapsto -\eta$ do preserve it. All four extend to the normalization (IC16), where their action is an independent sign on the labelled quadratic factors. This conclusion concerns those four source deck maps, not all automorphisms of a specialized split algebra.

7. LITERAL CHANNEL INTERPOLATION AND RECIPROCAL INVERSION

Suppose actual E and M states are both supplied at the same p . Denote their ordered literal roots by x_i and y_i . The exact root-algebra maps are

$$(IC27) \quad F(T) = \sum_i y_i \frac{\prod_{j \neq i} (T - x_j)}{\prod_{j \neq i} (x_i - x_j)}, \quad G(T) = \sum_i x_i \frac{\prod_{j \neq i} (T - y_j)}{\prod_{j \neq i} (y_i - y_j)}.$$

Evaluation proves both inverse compositions on the respective quotient rings. Under their common label coordinates in K^4 , the integral orders obey

$$(IC28) \quad \mathcal{O}_M \subset \mathcal{O}_E, \quad [\mathcal{O}_E : \mathcal{O}_M] = p^2.$$

Indeed the three M cluster values are congruent modulo p , in particular at the E pair of labels $0, 3$. The already proved indices give the ratio. Thus F is integral. In the reverse interpolation, the summand at label 2 has unit output x_2 and a denominator of valuation two; the other cluster outputs are divisible by p . Consequently

$$(IC29) \quad \boxed{\min_j v_p(G_j) = -2, \quad p^2 G(T) \bmod p = c_0 T^2 (T - a_M), \quad c_0 \neq 0.}$$

This proves the exact lost factor, not just an upper denominator estimate. At $p = 1009$, the pair of $(a, u) = (253, 11)$ states in (IC15) gives $c_0 = 306$ and $p^2 G \equiv 275T^2 + 306T^3 \pmod{1009}$.

The full signed generic algebras need not be isomorphic over K : E has ramified field factors and M has none. The surviving comparison is explicit after a specified extension. Adjoin square roots $\delta_i^2 = d_i^M/d_i^E$ and send η_M to $\delta(T)\eta_E$, where δ is the labelled interpolation of the δ_i . The inverse uses their reciprocals. The extension has degree at most four: for odd p , unit classes are square or a fixed nonsquare unit, and the only additional parity is that of p . The original integral images must still be tested by (IC17); they are not made unit lattices by this field isomorphism.

This is applicable to actual source-changing returns, for example the preceding capacity certificate [7, sharp finite repair] at $p = 825241$:

$$E : (a, R, u, h, r, s, \kappa) = (206321, 43, 2240439739, 19, 10859, 1, 208402140),$$

$$M : (a, R, u, h, r, s, \lambda) = (217170, 43439, 25, 1, 5, 43434, 1).$$

The latter is supplied there by the noncanonical original cofactor word 25, not by interpolation in (IC27). The current certificate transports both actual states through (IC18)–(IC29), preserving their different factor inventories. No M state is assumed to exist in order to use (IC27) as an existence proof.

7.1. The reciprocal model changes the lattice and the signed square class. Keep all four roots while applying $z_i = p/\ell_i$. On the literal root algebra,

$$(IC30) \quad I(T) = -\frac{p}{D}(AT^3 + T^2 + BT + C) = p/T.$$

Its exact coefficient denominator is p in E and p^2 in M. The transformed monic polynomial is $Z^4 L(p/Z)/L(0)$, where $L = \prod(T - \ell_i)$. If $N = \prod_i \ell_i$ and the normalized reciprocal quartic is $\tilde{\mathcal{H}} = -\frac{1}{5} \prod(Z - z_i)$, differentiation gives

$$(IC31) \quad \tilde{\mathcal{H}}'(z_i) = \omega \frac{\mathcal{H}'(\ell_i)}{\ell_i^2}, \quad \omega = -\frac{p^3 S}{5N}.$$

The four reciprocal roots sum to five; the coefficient $-1/5$ is therefore fixed, not fitted. The signed comparison needs the recorded quadratic twist $\sqrt{\omega}$. Its valuation is one in E and zero in M.

The reciprocal discriminant valuation is the literal value plus $12 - 6v_p(N)$, hence two in E and zero in M. The reciprocal root order has index p in E and is maximal in M. In E its colliding labels are now 1, 2, not 0, 3. Removing the separately marked root $z_0 = 1$ yields the preceding reciprocal cubic, whose signed derivatives acquire the further factors $-(z_i - 1)/5$ from the augmented quartic. This states the maps and square classes rather than conflating the earlier seven-point and current eight-point fibres.

8. A SINGLE MISSING PRIME PASSES ANY FIXED FINITE INTEGRALITY WINDOW

The supplied real separation estimates [5, §1.2, equations (2)–(4)] are necessary at an integral witness. The next result gives a precise scope test for treating them, together with a finite local completion, as sufficient. It retains both numerical channel gates and the entire allowed numerical range of the proposed integer u . The one failed condition is original square-divisor availability, at one explicitly designated prime.

Theorem 8.1 (A joint, single-prime source obstruction). *For every fixed finite set of rational primes Σ , there are infinitely many hard primes p and integers a, R, u such that*

$$(IC32) \quad p/4 < a < p/2, \quad R = 4a - p = 3, \quad 0 < u < a^2, \quad R \mid 4u + 1, \quad R \mid u + a,$$

yet $u \nmid a^2$. There is just one extra prime occurrence in u , at a specified prime $l \notin \Sigma \cup \{p\}$. In particular

$$a^2/u \in \mathbb{Z}_q \quad (q \in \Sigma \cup \{p\}), \quad v_l(a^2/u) = -1.$$

The literal E and M rational denominator returns from these same numerical coordinates satisfy the ES identity and all the following conditions:

- (1) *their only nonintegral denominator has reduced denominator exactly l ;*
- (2) *at p they have the respective E and M local signatures, normalization indices, conductors and invariant factors proved above;*
- (3) *their four positive literal roots have successive gaps greater than one;*
- (4) *with $s_p = (p^2 + 3p)/4$ and $B_p = p + s_p(s_p + 1)$, both satisfy*

$$S < B_p, \quad \max(x, y, z) < s_p(s_p + 1)/3, \quad \text{Disc}(\mathcal{H}) \geq 144/B_p^6, \quad 2/S \leq |\mathcal{H}'(\ell_i)| \leq S^2;$$

- (5) *the same prime p has separately supplied original integer E and M states. The failed source in (IC32) is not an ES counterexample.*

Proof. Choose a prime $l \equiv 11 \pmod{12}$ outside Σ . Dirichlet's theorem [4, Theorem 18.1] permits this choice, and then supplies infinitely many primes in the reduced CRT progression

$$p \equiv 1 \pmod{840}, \quad p \equiv -1 \pmod{l}.$$

The moduli are coprime. Take $p > 4l$ and set

$$(IC33) \quad a = (p + 3)/4, \quad R = 3, \quad u = al, \quad h = a/l, \quad r = l, \quad s = 1, \quad \kappa = (pl + 1)/3, \quad \lambda = (l + 1)/3.$$

Both quotients are positive integers, and $l < a$ gives $0 < u < a^2$. Modulo three, $p \equiv a \equiv 1$ and $l \equiv 2$, so both gates hold. But $a \equiv 1/2 \pmod{l}$ and $\gcd(a, u) = a$. Thus $h = a/l$ is nonintegral with exact denominator l . Every prime of a occurs in u with its original exponent; the only extra coordinate is $v_l(u) = 1$, while $v_l(a) = 0$. In the centered original box this is exactly $\beta_l = 1$ outside $[0, 0]$. All the equations $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$ hold, but the positive rational grade h is not an allowed original integer grade.

The two explicit returns are

$$(IC34) \quad E_{\text{rat}} : (a, a\kappa/l, pa\kappa), \quad M_{\text{rat}} : (a, pa\lambda/l, pa\lambda).$$

Multiplication proves their reciprocal identities. Since p, a, κ, λ are all units at l , the middle displayed denominator has exact reduced denominator l and the other two are integers. Every declared finite place therefore passes the denominator test and the original divisor valuation test.

At the original prime, $p \nmid a, l, \kappa, \lambda$ and $(l/p) = (p/l) = (-1/l) = -1$. Also $a \equiv 3/4 \pmod{p}$ is a square. Thus the same unit and character calculations as Theorem 2.2 apply to (IC34): in the first return $4a\kappa \equiv 1$, while the second has normalized root product $a^2\lambda^2/l$ a nonsquare. The reduction patterns, local field alternatives and all integral normalization formulas follow exactly. This verifies local algebra for rational targets; it does not insert a missing integer factor into the original box.

For both triples write their largest denominator as $z = s_p q$, where $q = \kappa$ or λ and $s_p = pa$. Since $l < a$, $\kappa = (pl+1)/3 < (s_p+1)/3$, and the same bound holds for λ . Hence $z < s_p(s_p+1)/3$. Their other denominator is $z/(pl)$ in E and z/l in M. It is greater than $pa/3 > p+1$; also $p-a > 1$ and $z-y > 1$. Thus the ordered root list a, p, y, z has successive gaps greater than one, and $S = p+a+y+z < p+3z < B_p$. Four such roots have Vandermonde modulus at least twelve. The product of three differences at a selected root is at least two and at most S^3 . The claimed discriminant and derivative bounds follow from $\mathcal{H} = -S^{-1} \prod (T - \ell_i)$.

Finally retain the genuine endpoint data

$$R_E = l, \quad a_E = u_E = (p+l)/4, \quad (h_E, r_E, s_E, \kappa_E) = (a_E, 1, 1, (p+1)/l).$$

They are integral and return $(a_E, a_E\kappa_E, pa_E\kappa_E)$. There is also a genuine middle state: put $j = (l+1)/4$, $c = (p+1)/l$, then

$$(h_M, r_M, s_M, \lambda_M) = (j, 1, c, 1), \quad a_M = jc, \quad R_M = c+1, \quad u_M = j.$$

Its identity is $p = lc - 1 = 4jc - c - 1$. Since $l \geq 11$ and $c > 4$, both first-half inequalities hold. This gives the last assertion without assuming ES. $\square$

For $\Sigma = \{2, 3, 5, 7, 11\}$, choose $l = 23$ and the hard prime $p = 35281$. The numerical source is

$$a = 8821, \quad R = 3, \quad u = 202883, \quad (h, r, s, \lambda, \kappa) = (8821/23, 23, 1, 8, 270488).$$

The two nonintegral denominator triples are

$$E_{\text{rat}} = (8821, 2385974648/23, 84179571556088),$$

$$(IC35) \quad M_{\text{rat}} = (8821, 2489709608/23, 2489709608).$$

The E target has zero rational local signed points and the M target has four. Their local normalization and ramification types are different. The only forbidden original factor is $23 \mid u$, while $23 \nmid a = 8821$. A second complete example at $p = 54601$ is included in the certificate. Trial division proves these displayed primes.

This theorem does not concern all places simultaneously: the omitted place is explicit. Nor does it exclude an adaptive test of every possible denominator prime. It proves that a fixed finite collection of local observations, even with the original numerical ranges, both channel congruences and the stated real bounds, is not original square-divisor membership. For the broader arithmetic-approximation context, compare [2, Theorems 1.1, 1.8–1.9]. The separate companion fragment `crt_control_note.tex` preserves an additional exact CRT construction with three larger-prime control targets; those targets and their full equations are also independently checked.

9. VERIFICATION SCOPE, INFORMATION LOSS, AND CONTINUATION

The full replay covers every original first-half box for the 99 primes $p \equiv 1 \pmod{12}$ through 3000: 868359 square divisors and 4505 labelled states, of which 2451 are E and 2054 are oriented M. It compares all derivative valuations, characters, coefficient identities and local point alternatives. The deep lattice calculations expand every original normalization matrix at the four $p = 1009$ examples. The separate checker uses all minors for the invariant factors and independent Lagrange inversion, rather than the first program's elimination. Both test the direct p^2/p^3 boundary equations, the actual capacity crossing, and every supplied rational control.

The main verifier is standard-library Python. The separate implementation imports none of it or of the preceding work. Its original source enumeration is by primitive coprime r, s and $h = a/(rs)$, rather than by divisors of a^2 . The prime domains, original vectors, ordered states and finite computation ranges are all recorded in JSON. Checks of a finite prefix are not a proof that the original source is nonempty at every prime.

No source coefficient is created by normalization. The exact sequence $0 \rightarrow \hat{\mathcal{S}} \rightarrow \mathcal{N} \rightarrow \mathcal{N}/\hat{\mathcal{S}} \rightarrow 0$ retains all the finite torsion coordinates in (IC18)–(IC20). Taking a quotient, a rational extension, or a reduction modulo p is accompanied by its calculated kernel. A receiving zero is not relabelled as absence of an original divisor word, and an added nilpotent boundary point is not promoted to an original E/M word. The countermodels in Theorem 8.1 make that latter distinction quantitative.

The remaining Turn 7 task is still to force a positive original integer coefficient for each prescribed hard prime. The completed same-grade capacity classification remains available, and its actual source-changing maps are compatible with the lattice comparison here. Finite rank, properness, local normalization and the displayed approximation bounds do not supply the initial arithmetic state by themselves. No Turn 8 witness guarantee, universal ES proof, Lean build or independent mathematical review is claimed.

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- [6] The Clankers (2026), *ES–Fabel arithmetic bridge*, `es_turn7_fabel_bridge_20260920/core.tex`, §§2–5 and `root_orders.tex`. The reciprocal cubic and its seven-point chart are kept separate from the literal quartic used here.
- [7] The Clankers (2026), *Fixed-target capacity completion*, `es_turn7_capacity_completion_20260920/core.tex`, complete same-grade channel return, fixed-target classification, sharp finite repair, and original prime realization. The exact upstream files and hashes are listed in `source_reading.json`. No universal occupancy statement is imported.